EM/2023/110

EM/2023/043

State Design pattern:

Used when an object can be in different states, and it's behaviour changes depending on the current state.

A University is developing a smart parking management system to manage vehicles entering and leaving the campus. A parking session can move through the following statuses. 1. Available status. 2. Reserve state. 3. Occupied state 4. Payment due state. 5. Close state. The behaviour of the parking session depends on it's current state.

1. Explain why the state pattern is appropriate for this system? (4 marks)
2. Draw a state diagram for the smart parking management system? (4 marks)
3. Design a UML class diagram for the above? (6 marks).
4. Based on your UML diagrams write java implementation for the parking session context class? (3 marks).
5. The university later designs to integrate an existing legacy parking payment machine. the university can't modify the legacy class, because it's supplied by an external vendor. provide a solution using design pattern knowledge?

1. Because here parking session has different states like available,

13/07/2026. ~~Inclas~~ Quiz - OR. 3 IM/2021/043

An application supports 3 databases 1. MySQL 2. Postgre SQL 3 - Mongo DB. Developers currently create database objects as follows.

1. MySQL database db = new mysql database ( );
2. db.create (product);

this software architect suggest using the factory design pattern.

1. ~~In this the sub obj~~

1. What is the role of the database pattern?
2. State two advantages of using the factory design pattern in this scenario?
3. Why does the ~~delive~~ developer not need to know whether dB is ~~mySQL~~ mysql or postgresql.
4. How to ensure only one connected dB is active at a given time?

1. ~~To create and~~ manage ok classes and objects. x

Quiz - 05

IM/2023/043.

A social media application has a user profile class that stores basic information such as name, email and profile picture. overtime the app wants to add extra features to profile such as premium ~~bat~~ ~~match~~ batch, QFA, and profile theme. Instead of ~~changing the~~ modifying the user profile class directly / or ~~adding multiplying~~ creating multiple subclasses.

1/ What is the design pattern that can be used?

2/ Explain how the selected design pattern can be apply? (3 marks)

3/ Draw a class structure for the selected design pattern? (8 marks)

4/ Write java like code that shows how a user profile can we wrapped with multiple decorators? (3 marks)

~~5/ How can we~~

5/ discuss possible challenges when applying the selected design pattern in real world application as above?

6/ Consider aspects such as code complexity, debugging, performance & readability? (4 marks).

1/ Decorator pattern. ✓

2/ Here If we added multiple classes are added for those extra features the code will be become very large and sometimes we may not be use that ~~code explosion~~ may be happen due to add multiple

(4)

IM/2023/043.

university has introduced a smart printing system. Student from the computer lab, library, engineering lab & staff room can send print jobs to a single highspeed printer. A software developer creates a new printer manager object every time a department submits a print request. After few minutes university notices the following problems.

1. print jobs are process in a wrong order.
2. some print jobs are duplicated.
3. The printer occassionally becomes unresponsible.
4. Different departments display different printer statuses.

Q1 Explain 2 problems that could occur because multiple printer manager objects are created? (2 marks)

Q2. Recommend an appropriate design approach to solve this problem and explain why it is suitable? (3 marks).

CamScanner

6.5
11

Date No

IM/2023/043. Quiz 03. 27/07/2026.

A University library uses a library management system to manage book borrowing & returning. The existing system communicate with an all barcode Scanner using a following method .scan ( )

Soon, a library purchases a new barcode scanner from a different vendor. However the new scanner provides the following method read barcode( ) . The library management system contains hundreds of calls to scan book. The university doesn't want to modify the existing application

1. What would be the reason not to modify the existing code?
2. What would be the appropriate design pattern & explain the reasons?
3. Identify client, target, adapter, adaptee, ~~participations~~ participants in the solutions?
4. Draw an UML class diagrams showing the relationships among these participants?
5. What would be the approach to provide one simple ~~simple~~ interface For borrowing books by internally calling Inventory, payment, notification and login services.?

IM/2022/135

(13/20)

Online course completion system

An online learning platform offers different type of courses.

1. Programming course
2. Design course

When a student completes the course the system follows the same overall process

1. Validate the student course registration.
2. Deliver the course specific final assessments.
3. Evaluate the students performance
4. Generate a completion certificate.
5.

However, final assessment and evaluation process are different for each course.

Programming course.

- Students complete a coding challenge.
- The solution is evaluated using automated test.

Design Course

- Students submit a design portfolio
- The portfolio is evaluated by an instructor using an assessment rubric.

The company wants to ensure that all courses follow the same completion process by allowing each course type to customize the assessment and evaluation steps.

Q/

1. Identify the most appropriate design pattern and explain why this design pattern is suitable for the given scenario. (2)
2. Draw a UML class diagram for the solution (8)
3. Write Java line code for the abstract class. (3)
4. Write Java line code for programming course & design course (4)
5. The platform also allows students to access additional optional course features. They are course content, assignments and final assessment. Students may optionally add 1. AI learning assistent
   2. Premium learning materials & one to one mentor support.

Explain how to implement the given scenario. (UML diagrams & codes are not necessary) (3)

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mark by → IM/2022/049

A university is developing a smart parking management system to manage vehicals entering and leaving the campus. A parking session can move through the following sessions.

1) Available status
2) Reserve state
3) Pre Occupied State
4) Payment due state
5) Close State.

The behaviour of the parking session depends on its current state.

Q) 1) Explain why the state pattern is appropriate for this system. (4)

2) Draw a state diagram for the smart parking management system (4)

3) Design a UML class diagram for the above. (6)

4) Based on your UML diagram write java implementation for the parking session context class. (3)

5) The university later decide to integrate an existing legacy parking payment machine. The university can't modify the legacy class because it is supplied by an external vendor. Provide a solution using design pattern knowledge.

Answers

1) In the given scenario we have different parking statueses such as available, reserved, pre occupied, payment due and close state. Since the vehical behaviours are completely different in each state the suitable design pattern for this system is state design Pattern; Because it lets the objects (vehical) to change its behaviour completely depending on its current state without using so many chained if-else conditions.

2) State design pattern eliminates this 'if-else' block by turning each state into its own class.

IM / 2022 / 135

A software company is developing a technical support system to troubleshoot customer issues. When a customer submits a support request, it may need to pass through several levels of support. Levels are ; (1) L1 supports - Handles basic issues such as password resets & block accounts.

(2) L2 support - Handles technical issues such as software installation errors.

(3) L3 support - Handles database connectivity issues and performance bottlenecks.

(4) specialist support - Handles rare problems that can't be resolved from L1, L2 or L3.

Q) (1) Explain how the COR design pattern can be applied to the system. You should identify request, Handler and concrete Handlers. (4)

(2) Draw a UML class Diagram to represent the above system using the chain of Responsibility Pattern. (6)

(3) Write a java like implementation to demonstrate how request for database connectivity issue is process through a support chain. (6)

(4) Assume that every support level must follow the same overall procedure.

1) Log the support request.

2) Attempt to solve the issue.

3) Log the ~~implementation~~. Resolution.

However, each support level should be able to implement its own specific problem solving logic. Explain how you could design the system to enforce above procedure (4)

Answers

(1) Here the support request is gone through several layers till it get handled. When a customer submit the request, in L1 it is checked if the request can be handled related to basic password reset & block account issues. If not the request is passed to next level (L2) to check whether it can get handled there. As this the request go through a level by level until it gets handled by the correct level support. So the COR Design pattern allows the request to go through one level after

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9.5/11

A University Library uses a library management system to manage book borrowing & return. The existing system communicate with an old bar code scaner using the following method.

Scan book = 1

The library purchases a new bar code scaner from a different vendor. However the new scaner provides the following method.

Read bar code.

The library management system contains 100s of calls to scan book. The university does not want to modify the existing application.

01) What would be the reason not to modify the existing code.

02) What would be the appropriate design pattern & explain the reasons.

03) Identify client, target, adapter, adaptee, participants in the solution.

04) Draw & give UML class diagrams showing the relationships among these participants.

05) What would be the approach to provide one simple interface for borrowing books by internally calling inventory, payment, notification and login services.

Answers

01) The current systems can be used because of it's stability. So to use them we can use adapter design pattern without changing the existing code. ✓ 1

02). Adapter design pattern. ✓ 1

The library borrowing system earlier used the old bar code scanner while new system read bar code to communicate. Since the university doesn't want to modify the system, adapter design pattern will translate the format which bar code scanner send into what adaptee needs without modifying the code. Because of the incompatibilities in the system & the bar code reader, which don't have a same format, the adapter design patter translate the client data in to the formad what adaptee expect. ✓ 2

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IM/2022/135

14/15

A social media application has a user profile class, that choose basic information such as name, email, and profile picture. Over time the app wants to add extra features to profile such as premium batch, two factor authentication, and profile name. ① Instead of modifing the userprofile class directly or creating multiple sub-classes, what is the design pattern that can be used.

② Explain how the selected design pattern can be applied. 3

③ Draw a class structure for the selected design pattern. 8

④ Write Java like code that shows how the user profile can be wrapped with multiple decorators. 9

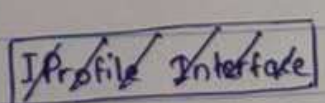
⑤ Discuss prossible challenges when appling the selected design pattern in real world applications. as above. Consider aspects such as code complexity, debugging, performance and readability. 4

Answers

① Decorator Design Pattern.

② According to above scenario the app requires to add two factor authentication, a premium batch & profile name to the same ~~user~~ profile ~~class~~ without changing that user profile class. By using decorator design pattern we can create seperate decorator cls for each additional feature like two factor authentication etc. and We can add premium batch cls decorator, authentication ~~decorator~~ profile class like wise decorater classes and make them they extend from the Profile Decorator class. So when user create those feature they will be created as the user wanted.

③



IM/2022/135

No: Date: / /

An application, supports 3 databases.

1. MYSQL
2. PostgreSQL
3. Mongo DB

Developers currently create database objects as follows.

MySQL database = SpaceDB = New ~~space~~ MySQLDB()

DB.create(product)*;

The software architect suggest using the factory designing patterns.

Q1) What is the role of the database

Q2) State 2 advantages of using the factory design pattern in this senario. 1/2

Q3) Why does the developer not need to know whether the database is MySQL or PostgreSQL

Q4) How to ensure only one DB connection is ~~activate~~ active at a given time ?

Answers

~~X~~ ~~Q1~~ * Q1) Store data

~~X~~ Q2) * Easy unit testing. - ~~Because products can be~~

* Improve code reusability & readability. - Objects are created in a centralized way so there is more cleaner, simple, readable code. so reusable logic can be used by multiple parts of the application

* Loose coupling - so client doesn't have to know about concrete class.

~~Q3~~ Q3) The factory decides which objects to create. There is no need to client / ~~to~~ developer to know about that. Ex. Client Developer depends on factory not the concrete classess. So it is easy to extent. 1

So it doesn't need to know whether the DB is MySQL or PostgreSQL

Q4) ~~The factory decides~~ ~~The factory~~ The ~~main program~~ doesn't worry about

No: Date: / /

03.

IM/2022/135- W.A.S.R.Fernando.

Smart printing system Student from the computer lab, library, engineeri & staff room can send print jobs to a single high speed printer. A software developer creates a new printer manager object everytime department submits a print request. After few weeks university notices the following problems.

1. Print jobs are processed in the wrong order.
2. Some print jobs are duplicated.
3. The printer occationally becomes unresponsive.
4. Different departments display different printer statuses.

Questions.

01) Explain two problems that could occur because multiple printer managers objects are being created. 2 m

02) Recommend an appropriate design approach to solve this problem and explain why it is suitable. 1 m

Answers

01) * There are going to be so many printer manager requests due to the current state of the printer manager. So the printer must have to operate for many requests at once. Because several departments may send requests at the same time.

* Due to the multiple printer managers the printer may not be able to operate correctly and in a correct order to print. Because So there gonna be missed prints & duplicated prints. This will cause inefficiencies in the printing process.

2